# SAHIL JHAWAR

sahil.jhawar448@gmail.com ◇ jhawarji ◇ sahiljhawar ◇ sahiljhawar.in

## EDUCATION

**Universität Potsdam** *October 2021 - June 2025*
Master of Science in Astrophysics *Potsdam, Germany*
Grade*: 1.7 (gut/good)
Thesis

**Christ (Deemed to be University)** *July 2018 - June 2021*
Bachelor of Science (Physics, Maths and Electronics) *Bengaluru, India*
GPA: 3.68/4.0

***Final Year Project***: Gesture Controlled System based on Raspberry Pi Pico *Report*

## WORK EXPERIENCE

**GFZ Helmholtz Centre for Geosciences** *July 2024 - Present*
*Research Scientist - February 2026 - Present* *Potsdam, Germany*
*Research Assistant - July 2025 - December 2025*
*Student Assistant - July 2024 - June 2025*
Software Developer in the Section 1.5 - Space Physics and Space Weather

- Developed a real-time ETL pipeline and testing framework for processing NASA/JAXA spacecraft data, implementing over 4,000 lines of code.
- Deployed and maintaining quasi-real-time space-physics based machine learning model and testing of the translated Python-based workflows.
- Migrated ∼2,500 lines of legacy code to Python, including data assimilation algorithms from MATLAB and geomagnetic models from FORTRAN.
- Collaborated with team members to ensure seamless implementation
- Developed software deliverables for the European Space Agency as part of funded research projects.
- Developed a Python-based space weather monitoring system using in-house forecasted Kp data for real-time analysis, threshold-based alerting, and automated email reporting.

**Leibniz-Institut für Astrophysik Potsdam (AIP)** *December 2022 - March 2024*
*Student Assistant* *Potsdam, Germany*
Student Assistant in the Project Management team for building (integration, installation and maintenance) the 4-metre Multi-Object Spectroscopic Telescope *4MOST*.

- Implemented *4MOST* commissioning procedures during 3 week coding campaign (November 2023)
- Utilized technical skills in troubleshooting and problem-solving within the specialized environment of an astronomical integration hall.
- Proactively managed procurement activities, liaising with vendors to procure essential large-scale items for the integration hall.

*See here on how to interpret German grade: Welche Form der Benotung wird an der Uni Potsdam verwendet?

## RESEARCH EXPERIENCE

**Theoretical Astrophysics Group, Universität Potsdam** *March 2023 - June 2025*
*Master's Thesis* *Potsdam, Germany*

Master's thesis student under **Prof. Dr. Tim Dietrich**.

- Maintaining and contributing to software infrastructure with over 60 forks, 40+ stars, and 15,000+ Conda downloads.
- Developed Bayesian models for analyzing astronomical datasets using NMMA, focusing on optimizing parameter estimation.
- Quantified and minimized non-stationary uncertainties by 2-10x in astrophysical parameter estimation and predictions.
- Designed a SHA-256 hash-based system to verify and automate file replacement for data consistency required for astrophysical simulations.
- Presented my findings as the first author on one research paper and as a co-author on another.

**Universität Potsdam & Astronomical Institute of CAS** *August 2022 - September 2022*
*Participant* *Prague, Czech Republic*

- Utilized the Perek 2m telescope at the Ondřejov Observatory to acquire stellar spectra
- Performed data reduction and analysis using IRAF Python for data visualization

**IUCAA & NCRA-TIFR** *December 2020 - January 2021*
*Winter Student* *Pune, India*

**Radio Astronomy Winter School 2020**: Learned basics of radio astronomy, from theory to instrumentation and computational aspects. Workshop enabled me to work on few physical experiments and on GMRT and ORT data of Vela pulsar.
*Detailed reports can be found here*

**Krittika, IIT Bombay** *May 2020 - August 2020*
*Summer Project Intern* *Mumbai, India*

Developed a Python package to analyse various types of binary star system.

## PUBLICATIONS


1. **Sahil Jhawar**, Thibeau Wouters, Peter T. H. Pang, et al. "Data-driven approach for modeling the temporal and spectral evolution of kilonova systematic uncertainties". In: *Phys. Rev. D* 111 (4 2025), p. 043046. DOI: 10.1103/PhysRevD.111.043046.
2. M M Desai, D Chatterjee, **S Jhawar**, et al. "Rapid parameter estimation for kilonovae using likelihood-free inference". In: *Monthly Notices of the Royal Astronomical Society* 541.3 (June 2025), pp. 2619–2630. DOI: 10.1093/mnras/staf1045.
3. T. Hussenot-Desenonges, T. Wouters, N. Guessoum, et al. *Multi-band analyses of the bright GRB 230812B and the associated SN2023pel.* 2023. arXiv: 2310.14310 [astro-ph.HE].


## OBSERVING EXPERIENCE

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|---|---|
| **Perek 2m Telescope** | Assisted with 7 nights (Ondřejov, Czech Republic) |
| **OST 0.5m Telescope** | Assisted with 2 nights (Potsdam, Germany) |

## TECHNICAL PROJECTS

**SWVO**
*contributor and maintainer*

Tools for downloading and reading space-weather data and geomagnetic indices.

**reuseify**
Automate REUSE license annotation from git history.

**Kp Alert**
Python-based system for monitoring forecasted Kp index and sending automated alerts.

**NMMA**
*contributor and former maintainer*
A Pythonic library for probing nuclear physics and cosmology with multimessenger analysis.

**nbody**
Direct N-body code with C++ and JAX backends.

**Gstrain**
Interactive Gravitational Wave strain viewer using Streamlit & Plotly. Check it out here

**BibQuest**
Automagically fill your bib file just using bib keys

**uptime**
A Github Action and Pages based uptime monitor in Python.

## SKILLS

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|---|---|
| **Technical Languages** | Python, C\C++, LaTeX, MATLAB, SQL |
| **Technical Skills** | Git (-Lab and -Hub), Unix based OS and Windows, Bayesian Inference, Data analysis and visualisation |
| **Packages and tools** | NumPy, SciPy, Pandas, Matplotlib, Astropy, Bilby, Seaborn, Sphinx, TensorFlow, scikit-learn, JAX, PyTorch, Streamlit, Selenium, bs4, PyBind, CI\CD, Parallel computing |

## OUTREACH AND COMMUNITY ENGAGEMENT

**Let's Talk Astronomy, Christ** *November 2019 - January 2021*
*Student coordinator* *Bengaluru, India*
Acted as a liaison between students and faculty. Conducted 15+ talks.

**LIGO-India, Vigyan Samagam (DAE-DST-NCSM, G.O.I)** *September and December 2019*
*Scientific Communicator* *Bengaluru and Kolkata, India*
Successfully played the role of science communicator by explaining GW Science and GW Exhibits to an audience of all ages. Catered to nearly 500 visitors.

## PRIZES, AWARDS AND HONORS

**2024** Fully funded visiting fellowship by European Union's Horizon 2020 Programme under the AHEAD 2020 hosted by Dr. Mattia Bulla at University of Ferrara, Italy
**2024** Fellowship of €500 by ESA for dotAstronomy 2024, ESAC Madrid, Spain
**2023** Scholarship of ₹35,000 by Badrilal Soni Maheshwari Shiksha Sahyog Kendra, India
**2021** Awarded co-curricular activities based scholarship from Christ (Deemed to be University)
**2021** Won numerous inter and intra collegiate events, and successfully lead a team of players resulting in various rolling trophies for the department
**2020** Gold honor in International Astronomy and Astrophysics Competition (IAAC), secured 38/40 and placed in the top 1% worldwide